

Product Requirements Document: ADHD-Friendly Study Support Tool

Background & Problem Statement

College students with ADHD often face unique challenges in managing coursework, particularly around procrastination and anxiety related to exams and assignments. ADHD is characterized by **executive dysfunction**, which impairs one's ability to initiate tasks, stay organized, manage time, and maintain focus ¹. This leads to an "intention-action gap" where students **know what to do but struggle to get started** ¹. Procrastination is a common result – not due to laziness, but often rooted in **fear of failure, perfectionism, or overwhelm** ². Students may wait for the "right" time or until panic sets in to start work, which erodes self-confidence and increases academic and emotional strain ³. Additionally, **test anxiety** can further hinder performance, especially if students resort to last-minute cramming or have had negative past exam experiences ⁴ ⁵.

Need/Opportunity: There is a need for a study system that bridges the intention-action gap and supports ADHD students in **turning intentions into actions** ¹. Such a system should help break down overwhelming workloads, instill structured routines, manage time more effectively, and provide coping mechanisms for anxiety. By leveraging technology (Canvas LMS data and AI) and research-backed strategies, we can create a personalized tool to improve focus, reduce procrastination, and alleviate anxiety for a junior CS student at Georgia Tech (our target persona) and others with similar needs.

User Profile (Target Persona)

- **Name:** (Example) Alex, 20 – Junior Computer Science major at Georgia Tech
- **Attributes:** Diagnosed with ADHD (combined type). Very intelligent and hands-on with learning, but struggles with **procrastination**, time management, and **anxiety** before exams/deadlines. Not currently on medication (tried non-stimulant without success), so relies on behavioral strategies.
- **Behaviors & Pain Points:** Tends to **put off large programming assignments** until close to the deadline, then experiences panic. Gets easily distracted when studying (social media, messages) and finds it hard to stay on one task. Has **inconsistent routines** – some days hyperfocuses on coding, other days can't get started at all. Feels overwhelmed by long to-do lists and often doesn't know how to prioritize. Before exams, Alex experiences high anxiety, fearing failure or blanking out, which worsens procrastination in studying.
- **Needs:** A structured yet flexible study plan that **prioritizes tasks** and breaks them into manageable chunks. Tools to minimize distractions and keep on track (timers, reminders). Techniques to make studying more engaging (interactive, multi-sensory methods) since passive reading doesn't hold attention. Support for building a **consistent routine** (including sleep, breaks, and exercise) and strategies to manage exam anxiety (e.g. gradual preparation, relaxation techniques). **Accountability** or motivation boosts (like rewards or a "study buddy" system) to initiate tasks and sustain effort.
- **Goals:** Stay on top of assignments **without last-minute cramming**, submit work on time, and prepare for exams in advance to reduce panic. Develop better study habits that leverage strengths

(e.g. hyperfocus on interesting topics) and mitigate weaknesses (e.g. losing track of time, difficulty starting tasks). Ultimately, improve academic performance and well-being by feeling more in control of schedule and stress.

Goals and Objectives

Primary Goal: Design a **personalized study support web application** that helps students like Alex (ADHD, prone to procrastination) manage their coursework proactively and reduce anxiety around deadlines and exams.

Key Objectives:

- **Proactive Task Management:** Encourage the student to start assignments and exam prep early by generating structured study plans. This combats last-minute rushing, a known trigger of anxiety ⁴, and replaces panic-driven efforts with steady progress.
- **ADHD-Friendly Strategies:** Integrate proven ADHD management techniques: breaking large tasks into smaller steps ⁶, using visual schedules and timers ⁷, and incorporating **frequent breaks** (e.g. Pomodoro technique) to maintain focus ⁸. The system should effectively **“design a system that reduces resistance”** for the ADHD brain ⁹, making it easier to translate intentions into actions.
- **Personalization & Canvas Integration:** Leverage the student’s real **Canvas LMS data** (courses, assignments, due dates, etc.) to create a plan that is **practical and tailored** to their actual workload. This ensures advice is context-aware (e.g., recognizing an upcoming midterm vs. a weekly quiz) and actionable.
- **Anxiety Reduction:** Incorporate guidance on **exam preparation and wellness** to alleviate anxiety. For example, recommend spreading studying over weeks (spacing), practicing with old exams, and self-care tips (sleep, exercise) which are shown to improve focus and reduce test anxiety ¹⁰ ¹¹.
- **Improved Time Management & Routine:** Help the student build a **structured weekly routine** that balances classes, study sessions, breaks, and personal time. Emphasize consistency (e.g., regular study times) while allowing flexibility for the unpredictable nature of ADHD ¹² ¹³. Over time, this should strengthen time-management habits and reduce the chaos that leads to missed deadlines.
- **Engagement and Motivation:** Keep the student engaged using **multi-modal and interactive techniques**. Leverage the student’s hands-on learning preference by suggesting active study methods (like quizzes, flashcards, or coding challenges) rather than passive reading. Implement motivational features (reward systems, gamification, or encouragement messages) since ADHD brains respond well to **short-term incentives and rewards**.

Success for this product means ADHD students can report: improved on-time assignment completion, less cramming, lower stress before exams, and a greater sense of control over their study process.

Solution Overview

We propose a **web-based multi-modal study planning application** integrated with Georgia Tech’s Canvas LMS and powered by a Large Language Model (LLM) for personalized guidance. The tool, tentatively called **StudyBuddy ADHD+**, will automatically import the student’s coursework data (assignments, quizzes, exams with due dates and estimated times) and generate a structured study plan tailored to their needs.

The core of the solution is an AI-driven planner that produces **clear, concise, and ADHD-friendly study plans**. The LLM (via a prompt) acts as an expert academic coach familiar with ADHD strategies, and it outputs a plan in organized Markdown (for easy readability). The plan will include: an overall strategy summary, prioritized focus areas, week-by-week study recommendations, exam prep tips, and time management insights – all customized to the student’s current workload and deadlines.

Multi-modal, interactive features are layered around this AI-generated plan to enhance execution: for example, an interactive calendar or timeline view of tasks (visual aid), the ability for the student to check off tasks (tactile engagement and dopamine reward), timers/alarms for work sessions (to enforce breaks and help focus), and perhaps optional text-to-speech or audio cues for those who prefer auditory reminders. The app will also provide motivational nudges (like positive reinforcement messages or small rewards for completing tasks) and options to involve peers (like sharing plans with a friend or finding a “study buddy” for body-doubling accountability).

In essence, the solution combines **data-driven planning** with **behavioral techniques** to create a comprehensive study system. It’s not just about telling the student *what* to study and when, but also about *how* to study effectively given their ADHD – incorporating routines, environments, and methods that maximize focus and minimize stress.

Key Features and Requirements

Below are the major features of the StudyBuddy ADHD+ app, each informed by research on ADHD and effective study habits:

1. Canvas LMS Integration and Data Import

- **Description:** Connect to the student’s Canvas account to retrieve real-time data on courses, assignments, quizzes, exams, due dates, and any provided time estimates or course calendar events. This provides the factual backbone for the study plan (e.g., how many tasks are pending, which deadlines are urgent).
- **Justification:** Using actual coursework data ensures the plan is **personalized and relevant** ¹⁴. It saves the student time (no manual input of tasks) and reduces the risk of overlooking any obligation, which is crucial for students with organizational challenges.
- **Functional Requirements:** The system must securely connect via Canvas API, update task lists daily, and compile a summary (number of courses, total tasks, tasks due in 7 days, etc.). This summary feeds into the LLM prompt.
- **Acceptance Criteria:** After linking Canvas, the user can see an updated list of their courses and tasks. The data used in the AI-generated plan matches the actual course deadlines (verified by sampling a few due dates).

2. AI-Powered Study Plan Generation (LLM)

- **Description:** Utilize an LLM (like GPT-4) to generate a **study plan** based on the Canvas data *and the student’s profile*. The prompt (system + user message) will be crafted to produce a structured Markdown output with specific sections: Overall Strategy, Priority Focus Areas, Weekly Study Recommendations, Exam Preparation Tips, and Time Management Insights.

- **Improvements for ADHD:** The prompt will explicitly mention the student's ADHD, procrastination tendencies, and anxiety, instructing the AI to incorporate **ADHD-friendly advice**. For example, it should suggest breaking down tasks and using micro-deadlines ("*break large tasks into smaller, manageable chunks*" ⁶), scheduling short focused sessions with breaks (Pomodoro method) ⁸, and using multi-sensory study techniques (like flashcards, mind maps) to sustain interest ¹⁵ ¹⁶. It will also encourage **early exam prep with spaced repetition** (reviewing material at intervals) since that improves long-term retention and is superior to cramming ¹⁷. Additionally, advice on mitigating anxiety (like practice tests, calming strategies) will be included if relevant.
- **System Prompt (Revised):**

```
export const STUDY_PLAN_SYSTEM_MESSAGE = `
You are an expert academic coach who specializes in helping college students
with ADHD
create effective study plans. You provide specific, actionable advice tailored
to their coursework
and mindful of ADHD-related challenges (procrastination, focus difficulties,
anxiety).
`;
```

This system message positions the AI as having domain expertise in ADHD coaching, ensuring the tone is empathetic and the advice is evidence-based for that context.

- **User Prompt (Revised Template):** The user prompt will incorporate a **Student Profile** section derived from onboarding or user input. For example:

```
**Student Profile:**
- ADHD (struggles with procrastination & focus, not on medication)
- Tends to feel anxious before exams/deadlines
- Prefers hands-on and visual learning methods

**Student's Current Workload:**
- Total courses: ${courseMapSize}
- Total tasks: ${tasksLength}
- Urgent tasks (due < 7 days): ${urgentTasksLength}

**Course Breakdown:**
${coursesSummary.map(course => `...`).join('\n')}

Please provide:
1. **Overall Study Strategy** ...
2. **Priority Focus Areas** ...
3. **Weekly Study Recommendations** ...
4. **Exam Preparation Tips** ...
5. **Time Management & Focus Tips** ...

(Keep the advice concise, actionable, in markdown.)`
```

(For brevity, the full template code is not shown, but the main change is adding the Student Profile and tweaking section 5 to explicitly mention focus tips.)

- **Functional Requirements:** The app must generate this prompt string with actual data and send it to the LLM API. The response (Markdown text) is then displayed to the user. The LLM output should be parsed and rendered with proper formatting (headings, lists, etc.).

- **Acceptance Criteria:** Given a test dataset of tasks, the AI returns a plan that **acknowledges the ADHD context** (e.g., it suggests pomodoros, breaking tasks down, using a planner or timers) and references the actual courses and deadlines appropriately. The plan should have all five requested sections, each with relevant content. No generic or irrelevant advice should appear – it should feel tailored to the provided data and ADHD needs.

3. Task Breakdown & Prioritization Engine

- **Description:** Provide functionality to break complex tasks (like projects or exam study material) into **smaller sub-tasks** within the app. For each Canvas assignment or exam, the tool can suggest or allow the user to create a checklist of micro-steps. For example, a term paper can be split into: research sources, write outline, draft introduction, etc., each with its own mini-deadline. The AI plan itself will recommend this approach (e.g., “divide your project into research → outline → code implementation → testing” in the Weekly Recommendations). The app should support it by letting users see and check off these sub-tasks.
- **Justification:** Breaking tasks into bite-sized pieces makes them **less overwhelming and more actionable** ⁶. It also creates momentum through small wins – *completing each sub-task provides a dopamine boost and sense of progress*, which is especially motivating for ADHD students ¹⁸. This directly addresses procrastination, as the student is more likely to start on a small, concrete step than a vague large task. Research shows this approach can reduce anxiety and prevent procrastination by making big projects feel manageable ¹⁹.
- **Functional Requirements:** When a new task is imported from Canvas, if it’s an “Assignment” or “Project” type, the app prompts the user (or auto-suggests via AI) to create sub-tasks. The user can edit these or accept suggestions. Sub-tasks can be assigned due dates prior to the main due date. The priority of tasks should be indicated (e.g., urgent tasks highlighted). The Priority Focus section of the AI plan will list which tasks need immediate attention and why – the app UI should reflect the same priorities (perhaps with an “urgent” label or a sorting by due date).
- **Acceptance Criteria:** Users can view any large assignment and see a breakdown of steps. They can check off steps as they complete them. The app clearly indicates which tasks or courses are high priority (e.g., an exam tomorrow would be flagged urgent). In user testing, students report the sub-tasks help them start working sooner and reduce feeling overwhelmed.

4. Structured Scheduling & Calendar Integration

- **Description:** The application will include a **calendar or timeline** view that schedules study sessions, leveraging the **time-blocking** strategy. Based on the AI’s recommendations and the student’s daily routine, the app can populate a study schedule. For example, if the AI suggests “allocate 5 hours this week to Algorithms coursework,” the app might schedule 1-hour blocks on available evenings and flag them as “Algorithms Study Session.” It will also incorporate **regular breaks** and downtime into the schedule, following the principle that frequent short breaks sustain ADHD focus ²⁰ ²¹. The schedule can be color-coded by course or activity type (e.g., blue for classes, green for study, yellow for breaks) ²² ²³, providing a visual structure.

- **Justification:** ADHD students benefit greatly from **visual aids and a clear routine**. A consistent schedule with set times for study, meals, and breaks improves focus and time management ²⁴ ²⁵ . Visual calendars and color-coding make it easier to follow the plan and not lose track of time ²⁶ . By scheduling tasks (with notifications/alarms), we reduce the reliance on the student to remember or decide when to study – it becomes part of their daily flow. This also helps prevent all-nighters; e.g., the app can ensure that exam prep is distributed in the calendar leading up to an exam (implementing *spaced study* to avoid cramming ¹⁰).
- **Functional Requirements:** Integrate with a calendar system (could be Google Calendar or an in-app calendar). The user should be able to input their fixed commitments (class times, work, etc.) initially, and the app fills in study times around those. The user can adjust the proposed schedule (drag-and-drop events). The app should send reminders or alarms when a study session is about to start (“Time to focus on CS Algorithms for 25 minutes!”). It should also remind to take breaks. Possibly include a **timer widget** (for Pomodoro) that the user can start for a session and that will alert when break time is up.
- **Acceptance Criteria:** The student’s week view clearly shows all classes, study sessions, and break times. Each study session entry corresponds to a specific course or task (and ideally ties back to the tasks list). The student receives timely reminders. In practice, users should feel the calendar is *helping* rather than rigid – they can report that it was easy to follow and adjust the schedule. Over a few weeks of use, the student should start developing a habitual routine (e.g., always studying at certain times as suggested).

5. Focus & Distraction Management Tools

- **Description:** To address distractibility, the app will include a **“Focus Mode.”** When activated, Focus Mode could do things like: fullscreen the study task at hand, play background white noise or binaural focus music, and (if possible) temporarily block distracting apps or websites on the study device (perhaps via browser extension). It will also encourage use of **noise-canceling headphones and a dedicated study space**, as noted in the plan and tips ²⁷ ²⁸ . The interface can display a checklist or “parking lot” notepad for intrusive thoughts – if the user suddenly remembers a different task or idea, they can jot it in the app’s notepad, then get back to studying (mirroring the “parking lot” technique ²⁹). After the focus session, they can review those notes.
- **Justification:** ADHD isn’t a deficit of attention but **an abundance of attention without control** ³⁰ . Reducing external distractions (via environment control and tech aids) is crucial to help the student channel their attention to one task ³¹ . Tools like site blockers, timers, and a minimalist interface can significantly improve productivity by keeping the student on task ³² . Meanwhile, capturing stray thoughts in a parking lot notepad prevents derailing (“I’ll quickly check this”) while reassuring the student they won’t forget that thought ²⁹ . These strategies directly support better focus and task initiation for ADHD users.
- **Functional Requirements:** Focus Mode should offer one-click activation. If possible, integration with known website blocker APIs or guidance to manually set up (for instance, instructing the user to use an app like Forest or a Chrome extension, possibly triggered by our app). The notepad should be simple and reset or archive after each session. Optionally, allow the user to customize background sound or ambient visuals when in Focus Mode to increase stimulation (some may prefer white noise ³³ or a visual timer).
- **Acceptance Criteria:** Users who activate Focus Mode manage to study without checking social media or other sites for the set duration. In surveys, users report fewer distractions and greater flow. The presence of a notepad for stray thoughts is utilized and appreciated (users feel it helps “clear their mind” of worry about other tasks).

6. Multi-Modal Learning Aids

- **Description:** The app will encourage and facilitate **multi-modal learning techniques** to keep study sessions engaging. For instance, it can generate or prompt the creation of **flashcards** for key concepts, or suggest the student do a quick quiz after reading notes. It might allow the student to upload or draw **mind maps** for their courses, or even provide a simple sketch pad to diagram concepts (especially useful for visual thinkers). The app's AI could assist by converting student notes or lecture slides into Q&A flashcards or summary glossaries. Audio features might include letting the student record short voice memos (e.g., explaining a concept in their own words, following the Feynman technique) and then listening to them while on a walk. These artifacts (flashcards, mind maps, voice notes) would be linked to tasks or courses and scheduled for review in later study sessions (implementing spaced repetition).
- **Justification:** ADHD brains **thrive on stimulation and novelty**, so varying the study methods is beneficial ¹⁵ ³⁴. Passive reading quickly becomes boring ³⁵, whereas active techniques like flashcards and self-quizzing promote deeper understanding and longer retention ³⁶ ¹⁶. **Multi-modal learning**, such as highlighting, doodling, or listening to recordings, can be highly effective for ADHD students ³⁷. It engages multiple senses and keeps the brain interested. For example, creating a mind map engages visual-spatial skills and forces the student to organize information hierarchically, which improves understanding. Spaced repetition (reviewing over increasing intervals) is proven to improve memory and problem-solving ¹⁶, and is especially important for students who might otherwise cram and forget. By building these techniques into the app (or at least reminding the student to use them), we align the tool with research-based study practices.
- **Functional Requirements:** Include a flashcard module (or integrate with existing apps like Anki via export/import). Provide a canvas for mind mapping or note doodling. Ensure the AI's plan mentions these strategies when relevant (e.g., "use flashcards for CS algorithms definitions"). Possibly incorporate an **AI tutor** mode: the student could quiz themselves by asking the LLM questions or having it ask them practice questions (limited by what data we have access to – perhaps user-provided notes or generic concept questions if not violating academic integrity). The app should allow scheduling of these activities (e.g., "Thursday: review flashcards from Chapter 3").
- **Acceptance Criteria:** Students use at least one additional modality beyond reading in their study sessions (as tracked by usage of the flashcard or mind map features). Feedback indicates the variety helps maintain interest. For example, a student might say they felt less bored and more engaged by switching between reading, quizzing, and drawing out ideas, which is the intended outcome.

7. Motivation and Gamification Features

- **Description:** To keep students motivated, the app will incorporate **gamification and reward mechanisms**. This could include: a reward points system for completing tasks or sticking to the schedule, unlocking badges/achievements for streaks (e.g., "Studied 5 days this week!"), or integrating with habit-building apps like Habitica/Forest via API. The app can also prompt the student to **set a reward for themselves** for meeting a goal (like "After finishing your math homework, reward yourself with 30 minutes of gaming" – the AI can suggest such rewards as it knows the student's interests). We will also include an option for **accountability partnerships**: e.g., the app can share progress updates with a chosen friend or allow two friends to connect and view each other's task completion (with consent), similar to a social study group feature. Even a simple **"share your plan"** link could let the student show someone (advisor, tutor, or parent) what their plan is, creating external accountability.

- **Justification:** ADHD students often need stronger, more immediate incentives to initiate and sustain effort. Gamification leverages this by providing instant feedback and a sense of achievement for small steps. Research and anecdotal evidence suggest that **short-term rewards** can significantly boost motivation in ADHD brains. Additionally, having a supportive community or a study buddy creates accountability and moral support ³⁸ ³⁹ . Body doubling (working alongside someone) or just sharing goals with a friend can increase follow-through ⁴⁰ ⁴¹ . By making the study process a bit more game-like and social, we address the motivational deficit and make studying feel less like a solitary chore.
- **Functional Requirements:** Points/badges should be visually represented in the UI (a dashboard showing daily points, etc.). The app might include a leaderboard if used in a group setting (opt-in). The reward suggestion feature can be part of the AI output or a manual input where the user sets a promised reward. Social features require user accounts and the ability to find/add friends or share links securely. All gamification elements should be optional and non-distracting (the student can silence them during focus time).
- **Acceptance Criteria:** Users who engage with the gamification report higher motivation – e.g., they complete more tasks per week compared to baseline. A/B testing could show that those with rewards enabled stick to their plan longer. If an accountability buddy is used, the user meets deadlines more consistently (since someone else is aware of their goals). Qualitatively, students should find the rewards fun and not childish, and feel that it helps them push through tasks.

8. Stress Reduction & Exam Anxiety Support

- **Description:** Beyond scheduling and task management, the tool will provide resources and prompts for **managing anxiety**. This includes a library of quick **stress-reduction exercises** (like guided deep breathing, short meditations, or even a 5-minute “stretch break” video) that the app can recommend during breaks or when it detects the student is feeling anxious (the student might press an “I feel anxious” button, prompting a coping tip). The AI’s “Exam Preparation Tips” section will also address anxiety by suggesting best practices: e.g., *“Start reviewing at least 2 weeks before the exam using spaced repetition ¹⁰ , and do practice problems to build confidence. Get a good night’s sleep before the exam and avoid last-minute cramming ⁴² .”* The app can send **nudges in advance of exams**: “Your exam is in 3 days, how about a quick review session? Remember to prepare your materials and avoid caffeine overload ⁴³ .” We can also integrate an interface for the student to create a personalized **“calming playlist or worksheet”** (as UNC suggests) with motivational quotes or images of supporters ⁴⁴ to look at when stressed.
- **Justification:** Anxiety can paralyze students and undo their preparation. By addressing it directly, we improve academic outcomes and student well-being. Early and effective studying is one antidote to test anxiety (because feeling prepared reduces fear of failure) ⁴⁵ ⁴⁶ . Additionally, mindfulness and relaxation techniques are proven to help manage acute anxiety symptoms ⁴⁷ ⁴⁸ . Incorporating these elements means the app isn’t just about *when* and *what* to study, but also *how to stay calm and confident* while studying and taking tests. This is especially crucial for ADHD students who might have had negative academic experiences and thus higher baseline anxiety.
- **Functional Requirements:** Include a “Wellness” or “Anxiety Relief” section with exercises (possibly embed YouTube videos for guided breathing, or text instructions for progressive muscle relaxation). The app should detect high-stress periods (like the night before an exam in the schedule) and proactively offer support (“It looks like tomorrow is a big day, here are some tips...”). Ensure that the AI’s plan always includes an Exam Tips section when an exam is upcoming, and that it provides both study strategies and mental preparation advice. Possibly integrate with campus resources: a link to

counseling services or disability services if the student feels they need more help (the plan or app can remind that reaching out is okay ⁴⁹ ⁵⁰).

- **Acceptance Criteria:** Students report feeling less anxious because they followed the plan (e.g., “I started studying early and knew what to expect, so I wasn’t as nervous”). In-app survey after exams could measure anxiety levels vs. prior exams. Usage stats should show students actually engaging with the stress-relief tools (e.g., X% of users played a breathing exercise before an exam).

9. Personalization & Adaptivity

- **Description:** Over time, the app should adapt to the student’s patterns. For example, it might learn the student’s most **productive times of day** (if Alex focuses best in the morning, schedule important tasks then) ⁷ . It can also note if the student consistently misses a certain scheduled study time and adjust the routine or send a gentle reminder to review the schedule each week ¹² . The AI prompt can be regenerated weekly (or on demand) to update the plan based on new tasks or tasks not completed from last week. The system should allow the student to give feedback (like “this advice was helpful” or “too much work on Tuesday”) to fine-tune future recommendations.
- **Justification:** ADHD management is not one-size-fits-all; it often requires **regular review and adjustment** of strategies ¹² . By personalizing the schedule to what actually works for the individual and staying flexible, we increase the likelihood of long-term adherence. The student will feel the tool “gets them,” rather than imposing a rigid plan. This adaptive approach is supported by coaching best practices – academic coaches often have students reflect weekly and adjust plans accordingly ⁵¹ .
- **Functional Requirements:** Track user behavior (with consent) such as task completion times, which study sessions were skipped, etc. Use simple heuristics or AI to suggest changes (e.g., “We noticed you focus better in 30-minute sessions than 60-minute ones, shall we adjust your default Pomodoro length to 30?”). Provide an easy way to regenerate the weekly plan (maybe every Sunday evening, as Zeam’s strategy suggests ⁵²) incorporating any new tasks. Ensure data privacy in tracking.
- **Acceptance Criteria:** After a month of use, the student’s schedule is more optimized (fewer missed sessions, a balance that fits their life). The student feels improvements in productivity and reports that the app’s suggestions became more helpful over time as it “learned” their preferences.

Non-Functional Requirements

- **Usability & UX:** The interface must be clean, calming, and not overwhelming. Information should be broken into small, digestible pieces (much like the study tasks themselves). Use **short paragraphs and bullet points** in the AI’s output, per the prompt, to ensure readability. The app should support **Markdown rendering** for the plan to maintain formatting. Navigation should be intuitive, as ADHD users may get frustrated with complex menus. Provide quick access to key features (Plan, Calendar, Focus Mode, etc.).
- **Performance:** The app (and especially the AI response generation) should be reasonably fast. A delay of a few seconds for generating a plan is acceptable, but the rest of the UI (loading tasks, calendar) should feel snappy. Offline access to the plan (read-only) could be a plus, in case the student is offline but still wants to check their schedule.
- **Privacy & Data Security:** Integrate with Canvas via secure OAuth; store user’s academic data and any notes in an encrypted database. Since personal study habits and potentially health-related info (ADHD status) are involved, ensure compliance with privacy laws (FERPA for academic data, possibly HIPAA if we ever integrate medical info about ADHD – likely not, but treat it sensitively). No data is shared with third parties without consent.

- **Compatibility:** Web-based app should be accessible on all modern browsers. A responsive design is needed so that the student can check their plan on a mobile device as well. Perhaps consider a future mobile app for push notifications.
- **Accessibility:** Ensure the app is usable with common accessibility tools (screen readers, etc.), as some ADHD students also have co-occurring learning disabilities. Use clear fonts, an option for dyslexic-friendly font, adjustable text size, and dark mode for those who prefer it. Avoid excessive animations or clutter that could distract.
- **Multi-Modal Support:** Aside from the study multi-modal features, the app itself could allow **voice input or output** – e.g., the student could ask the app (via speech) “What’s my plan for today?” and get a spoken response. This would help on days when they have trouble getting started (they could use an Alexa-like interaction to initiate their study). While not core, this aligns with leveraging all modalities to help the user engage with the plan.

Competitive Analysis & Differentiation

(Briefly, to contextualize) There are existing study apps and ADHD planners (Notion templates, Todoist, Forrest, Pomodoro apps, etc.), and even some AI study plan generators. However, **StudyBuddy ADHD+** differentiates by combining all these functions into one cohesive tool tailored for ADHD needs. Unlike a generic planner, our tool explicitly addresses procrastination through AI coaching and breaks tasks down automatically. Unlike basic to-do apps, it integrates **Canvas data** and uses an LLM to provide rich, personalized guidance (almost like having a tutor or coach available 24/7). Many tools require the student to self-manage (which is exactly what ADHD impairs); our system offloads that executive function burden onto an AI assistant and structured system, letting the student focus on execution. This guided, research-backed, and comprehensive approach is novel in the ed-tech and productivity space.

Metrics for Success

- **Engagement:** Daily/weekly active use of the app (especially during the semester). High engagement would indicate the tool is becoming part of the student’s routine.
- **Task Completion:** Increase in the percentage of assignments completed *before* the due date (Canvas data can track submission times). We aim to see procrastination decrease (e.g., assignments done at least a day in advance).
- **Exam Preparation:** Survey or usage data indicating students start studying for exams earlier than they used to (perhaps compare self-reported habits from before using the app vs. after).
- **User Satisfaction:** High satisfaction ratings, especially regarding reduced stress and feeling more in control. Qualitative feedback like “I feel less overwhelmed” or “I managed to study consistently for my midterms and it paid off” would validate the approach.
- **Academic Performance:** As a longer-term metric, GPA or exam scores might improve. (Though many factors affect grades, a notable improvement in the targeted areas – e.g., fewer missed assignments, higher quiz scores from better studying – would be a positive sign.)
- **Retention:** The student continues to use the app across semesters, and perhaps even refers it to others (indicative of value).

Future Extensions

In future iterations, the app could incorporate more sophisticated AI tutoring (answering questions about course material), deeper integration with university support (linking to tutors or coaches), and community

features (group study sessions, forums for ADHD students to share tips). It could also adapt for other profiles (e.g., students with dyslexia or other learning differences) by tweaking the prompt and feature set accordingly.

By grounding the design in proven **pedagogical and behavioral strategies** for ADHD, and leveraging technology for automation and personalization, this study support tool aims to significantly improve the academic experience for students like Alex. It will help them **work smarter, build positive habits, and overcome procrastination and anxiety** through a structured yet flexible system of support – effectively acting as an always-available, research-informed academic coach in their pocket.

Sources:

- Amina A. Mohamed, *Maximizing Study Productivity for Students with ADHD* – Highlights the executive functioning challenges (intention-action gap) and strategies like environment design, time-blocking, micro-steps, and Pomodoro technique ¹⁴ ⁵³ ⁸ . Emphasizes designing systems around one’s “brain rhythms” and using external support ⁹ .
- Weingarten Center (UPenn), *Structured Schedule for Students with ADHD* – Recommends establishing routines, visual schedules (color-coding), breaking tasks into chunks, frequent breaks, and regular review of the schedule for adjustments ²² ⁶ ²⁰ ¹² . These informed our scheduling and adaptability features.
- ADD.org, *How to Study Efficiently with ADHD: 7 Tips* – Suggests multi-modal and active study techniques (flashcards, spaced repetition, Feynman, mind mapping) to keep the ADHD brain engaged ¹⁶ ⁵⁴ . Also advises minimizing distractions via tools and the “parking lot” method ³² ²⁹ , and using stimulation (colors, white noise, fidgeting) to improve focus ⁵⁵ ⁵⁶ . We incorporated these into the focus mode and multi-modal features. Additionally, it discusses motivation hacks like setting realistic goals, breaking tasks down, and gamification (Habitica, Forest) ⁵⁷ ⁵⁸ , plus accountability through body doubling ⁵⁹ ⁶⁰ and the importance of rewards for ADHD brains ⁶¹ – all of which shaped our motivation and social features.
- Zeam Health, *9 Strategies to Control ADHD in College* – Reiterates structured routine and Pomodoro breaks ⁶² , joining support groups for accountability ³⁸ , planning weekly and breaking tasks to reduce anxiety ⁶³ ¹⁹ , utilizing campus resources ⁶⁴ , optimizing study environment (quiet space or noise-cancelling) ⁶⁵ , healthy lifestyle (sleep, exercise) ⁶⁶ , and coping strategies like CBT and coaching ⁶⁷ . These points reinforced many of our features (routine, community, environment control, wellness).
- UNC Learning Center, *Tackling Test Anxiety* – Details causes of test anxiety (fear of failure, perfectionism, lack of preparation) and strategies to mitigate it: **start early, space out studying, self-care, arrive prepared, positive self-talk, and mindfulness during exams** ⁴⁵ ¹⁰ ⁶⁸ ⁴⁷ . This guided the exam prep and anxiety-relief aspects of our solution.

¹ ² ³ ⁷ ⁸ ⁹ ¹⁴ ¹⁸ ⁵³ Maximizing Study Productivity for Students with ADHD – Learning Services
<https://learningservices.gmu.edu/newsletter/maximizing-study-productivity-for-students-with-adhd/>

⁴ ⁵ ¹⁰ ¹¹ ⁴² ⁴³ ⁴⁴ ⁴⁵ ⁴⁶ ⁴⁷ ⁴⁸ ⁶⁸ Test Anxiety – Learning Center
<https://learningcenter.unc.edu/tips-and-tools/tackling-test-anxiety/>

6 12 13 20 21 22 23 24 25 26 27 28 51 How to Create a Structured Schedule: Strategies for College Students with ADHD – Weingarten Center

<https://weingartencenter.universitylife.upenn.edu/creating-a-structured-schedule-with-adhd/>

15 16 17 29 30 31 32 33 34 35 36 40 41 54 55 56 57 58 59 60 61 How to Study Efficiently with ADHD: 7 Tips to Boost Focus & Motivation

<https://add.org/tips-for-studying-with-adhd/>

19 37 38 39 49 50 52 62 63 64 65 66 67 9 Strategies to Help Successfully Control ADHD in College - Zeam Health & Wellness | Psychiatry, Mental Health, Primary Care & Aesthetics

<https://www.zeamhealth.com/9-strategies-to-help-successfully-control-adhd-in-college/>